ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Weather-Based Prediction of Wind Turbine Energy Output: A Next-Generation Approach to Renewable Energy Management

1. Introduction

Project Title: **Weather-Based Prediction of Wind Turbine Energy Output: A Next-Generation Approach to Renewable Energy Management**

2. Project Overview

• **Purpose:** The Internship Management System is a web-based application developed during the internship period.
The purpose of this project is to design and implement a full-stack application that allows users to register, log in, and manage records efficiently through a connected backend and database system.

Features:

User Registration and Login
o Authentication and Authorization
o Dashboard for managing records
o Create, Read, Update, Delete (CRUD) operations
o Database integration
o Responsive user interface

3. Architecture

• **Frontend:**

Built using HTML, CSS, and JavaScript
o Form validation implemented
o Dynamic content rendering
o Responsive design

• **Backend:**
Developed using (Node.js with Express / Flask – update based on your project)
o RESTful API structure
o Middleware for authentication and error handling

Database:

MySQL / MongoDB
o Tables/Collections:
▪ Users
▪ Internship Records / Data
▪ Admin (if applicable)

o Relationships:
▪ One-to-many (User → Records)

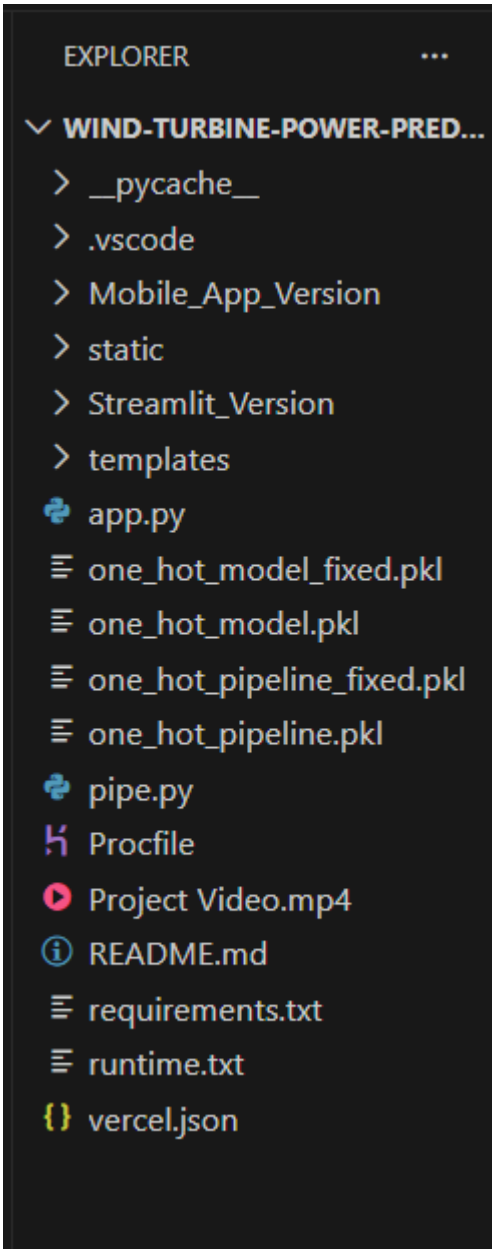
4. Setup Instructions

• **Prerequisites:** Node.js / Python
o npm / pip
o MySQL / MongoDB
o Git

• **Installation:**

1. Clone the repository
   git clone https://github.com/GayatriSatram3/internship
2. Navigate to project folder
   cd internship
3. Install dependencies
   npm install

5. Folder Structure



6. Running the Application

```
PS D:\Wind-Turbine-Power-Predictor___using-Machine-Learning-main> python app.py
https://scikit-learn.org/stable/model_persistence.html#security-maintainability-limitations
  warnings.warn(
C:\Users\user\AppData\Local\Programs\Python\Python310\lib\site-packages\sklearn\base.py:376: Inconsi
.1 when using version 1.4.2. This might lead to breaking code or invalid results. Use at your own ri
https://scikit-learn.org/stable/model_persistence.html#security-maintainability-limitations
  warnings.warn(
 * Serving Flask app 'app'
 * Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WS
 * Running on http://127.0.0.1:5000
Press CTRL+C to quit
 * Restarting with watchdog (windowsapi)
```

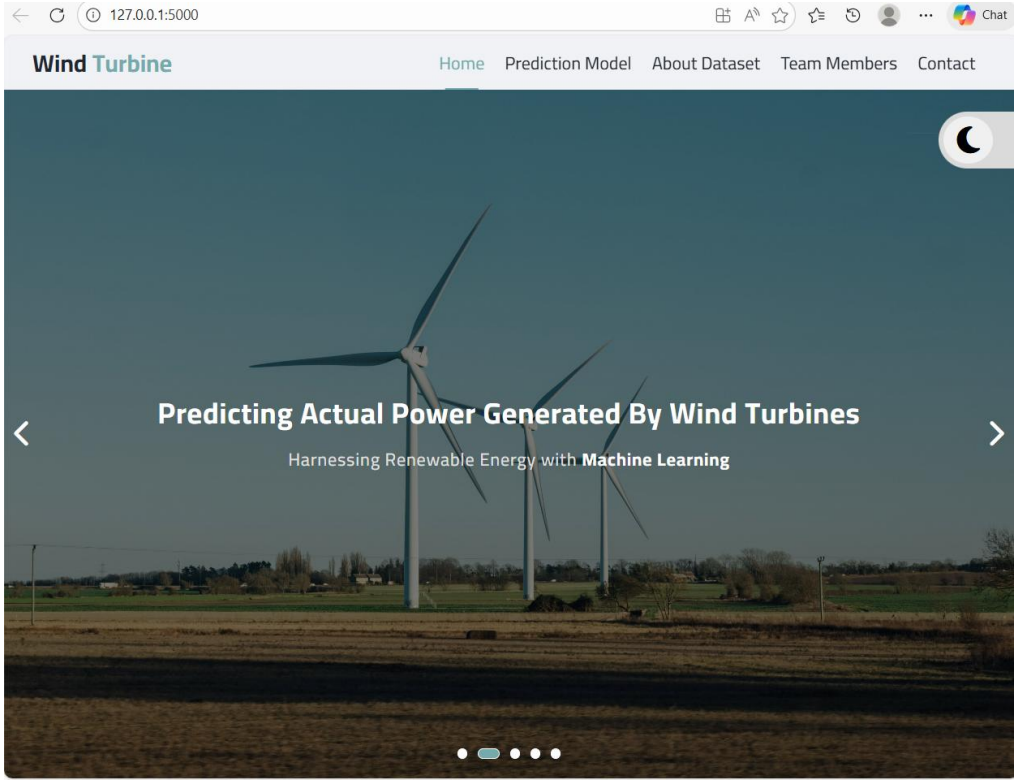
7. Authentication

• **Method Used:** JWT / Session-based Authentication

• **Workflow:**

o User logs in with credentials
o Token is generated
o Token is stored on client side
o Protected routes are verified using middleware

8. User Interface



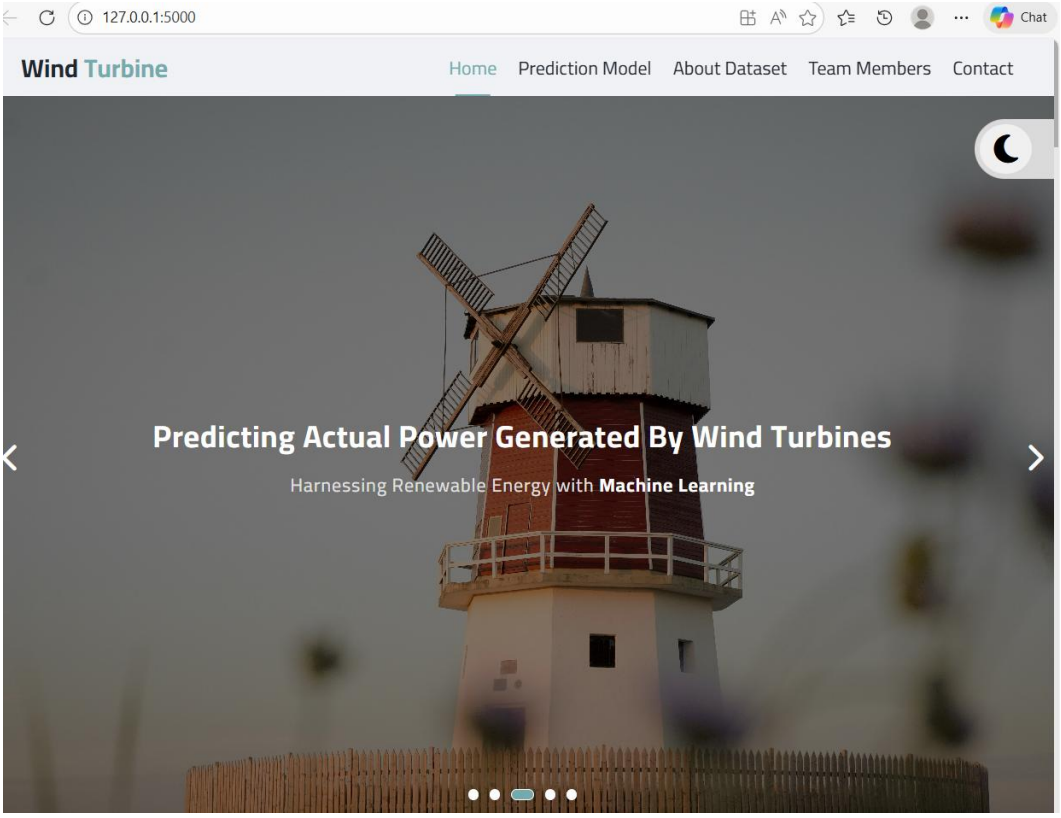
9. Testing

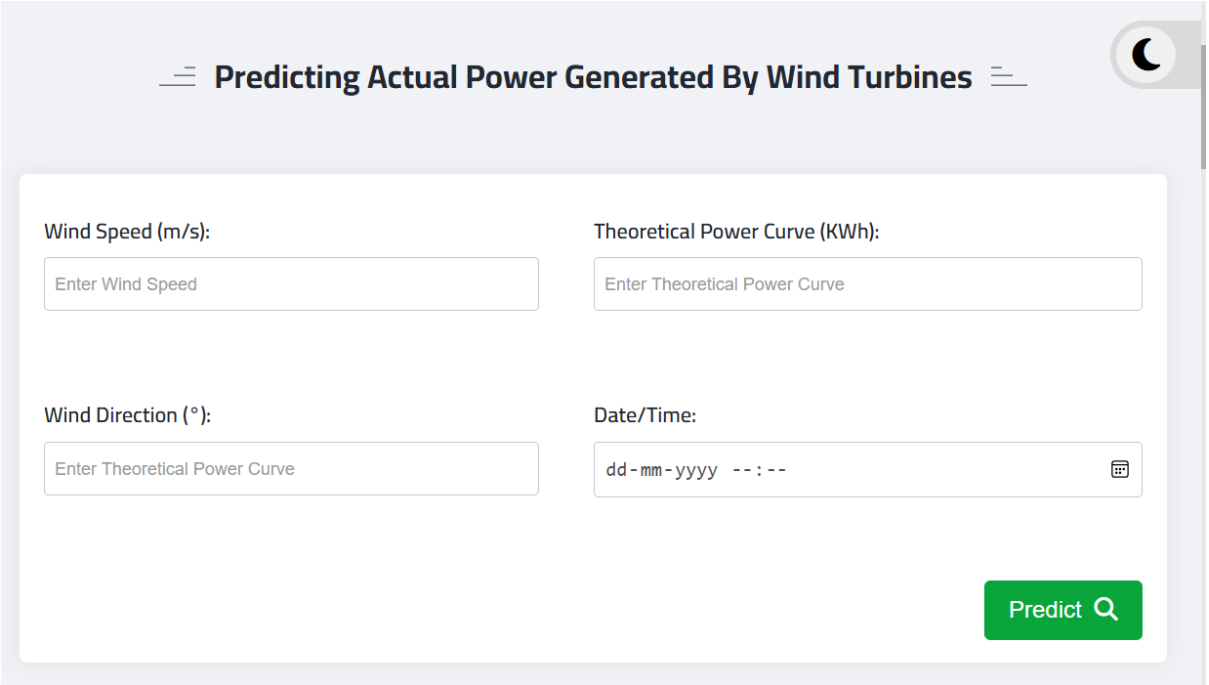
• **Tools Used:** Postman

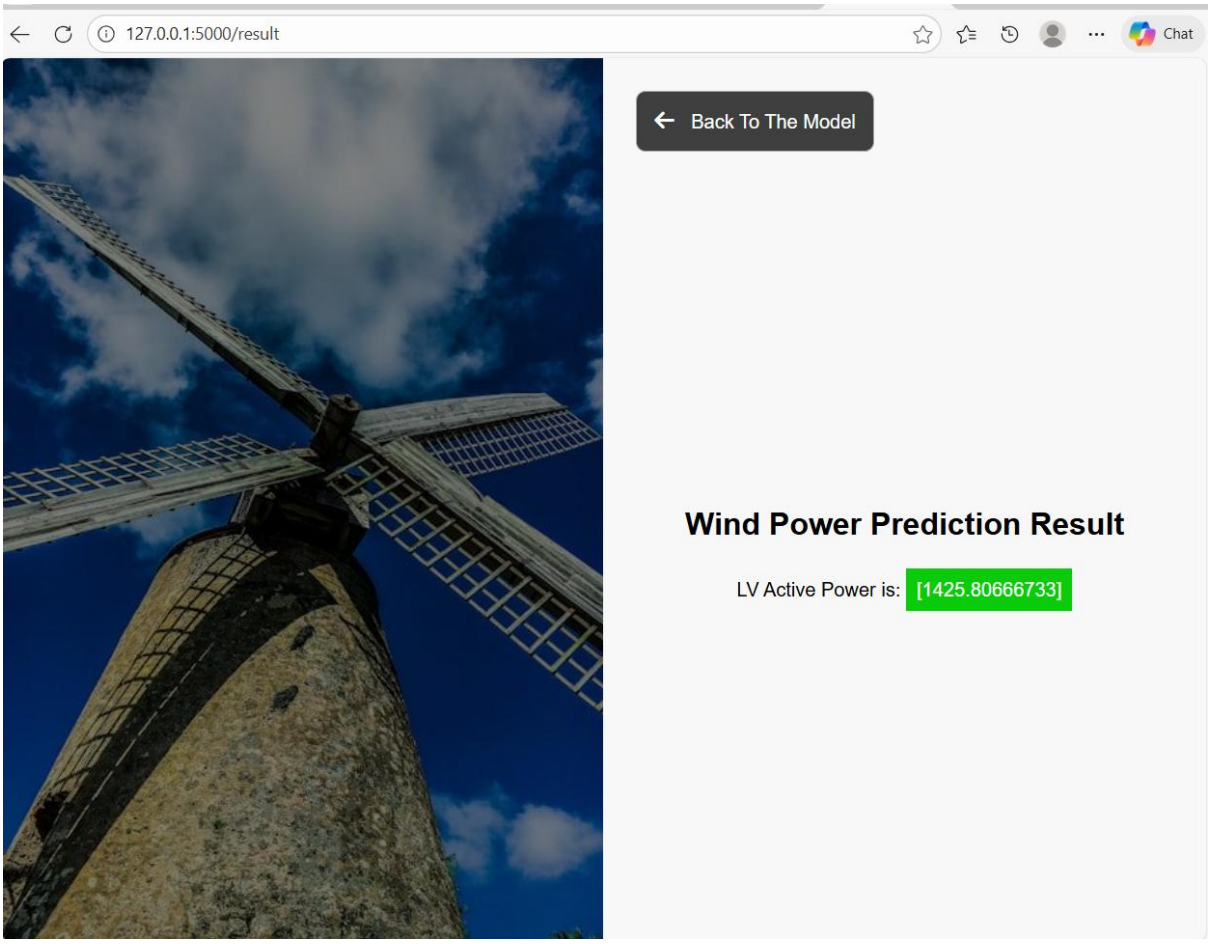
• **Strategy:**

o Manual frontend testing
o API testing using Postman
o Backend route validation
o Database connectivity testing

10. Screenshots or Demo







Known Issues

- No email verification system
- Limited automated testing
- UI can be further improved

Future Enhancements

- Role-based access control
- Email notification system
- Admin analytics dashboard
- Deployment on cloud platform
- Improved UI using modern frontend frameworks